

Overnight Basal Review

Andrew · Type 1 · Omnipod + Loop (closed-loop) · Dexcom G7 · **Data: Jun 24 - Aug 23, 2026**

Compiled from Nightscout CGM and pump records by a family member.
Observational — for clinical review, not a diagnosis.

The ask

Review the overnight basal rates (22:30 - 04:30). Loop is currently delivering far less basal than the schedule specifies — **~24.5 U/day actual vs 38.05 U/day scheduled** — and suspends 27–38% of the time overnight. The scheduled rate is what the pod falls back to if Loop stops (dead phone, lost pump comms), which is a meaningful concern now that he lives alone at college.

He has had **recurrent nocturnal severe lows** (nadirs of 40, 43, 50 mg/dL in the past week), all self-treated and resolved. Alarms, fast carbs and a roommate are in place.

A second item worth equal attention: the correction target is a **flat 100 mg/dL across all 24 hours**, and 81% of Loop's automatic overnight insulin is given below 150 mg/dL. **Raising the overnight target** may act more directly on the lows than the basal change, and requires no change in behaviour (\$4).

1 · SCHEDULED VS ACTUALLY DELIVERED BASAL

Hour	Scheduled	Loop Delivers	% Time Suspended	Mean BG	%<54
22:00	2.40	1.32	36%	120	0.3
23:00	2.20	1.28	38%	115	0.5
00:00	1.45	0.92	31%	113	0.5
01:00	1.45	0.88	33%	108	0.2
02:00	1.30	0.72	37%	106	1.3
03:00	1.30	0.90	27%	108	0.0
04:00	1.30	1.00	16%	112	0.0
05:00–08:00	1.35	1.01–1.11	10–15%	115–117	0.0

Delivered rates are reconstructed from Loop's temp-basal records; G7 compression artifacts excluded from the glucose columns. **Green rows (04:00–08:00) are the one part of the schedule Loop largely accepts** — it runs near the programmed rate, suspends only 10–16%, and produces no lows. **Red rows are where the schedule and reality diverge most.**

2 · PROPOSED CHANGE — OVERNIGHT ONLY

22:30 2.20 → **1.70**

00:00 1.45 → **1.20**

02:00 1.30 → **1.00**

04:30 1.35 → 1.35 (unchanged)

09:30 + all daytime unchanged

Total daily basal 38.05 → **36.05 U** (–5.3%)

All of the reduction between 22:30 and 04:30

Overnight vs Loop's actual +58–81% → **+30–39%**

A gentler variant (1.90 / 1.30 / 1.10, –1.25 U/day) is available if a smaller first step is preferred. Daytime is deliberately untouched so no new daytime highs are introduced — highs tend to prompt aggressive corrections, which is the pattern driving the overnight crashes (see \$4).

3 · IS THERE ROOM TO RUN HIGHER OVERNIGHT?

End of Night	Mean	P25	P75	%>180
07:00	117	100	132	0.6
08:00	115	97	130	0.8
09:00	113	95	124	2.4

Mornings currently end in the 113–117 range with essentially no time above 180, so a modestly higher overnight profile should not push him into morning hyperglycemia. **Expected effect on delivered insulin is small** — Loop is already suspending 27–38% overnight and will simply suspend less. The primary gain is the fallback rate and restored control headroom, not a large change in insulin actually given.

4 · BOLUS INSULIN OVERNIGHT — TWO SEPARATE SOURCES

The basal change above is a **safety and headroom** measure; on its own it is not expected to eliminate the lows. Boluses overnight come from two distinct sources, identified using Loop's own **automatic** flag on each dose (not inferred):

Overnight Window	Loop Automatic	Manual	Median BG at Dose
22:00 – 02:00	1.57 U	3.74 U	118 – 150
03:00 – 05:00	1.32 U	~0	120 – 126
Total per night	2.89 U	3.82 U	–

BG When Dosed (After Late Meals)	Loop Automatic	Manual
Under 120 mg/dL	29%	30%
120–150	52%	30%
150–180	17%	32%
180 or above	1%	8%

Two different problems. (1) **22:00–02:00 is mostly manual** — 3.74 U per night at a median BG of 118–150, i.e. dosing ahead of a rising trend rather than treating a high. Late-meal peaks are in fact modest (median 159; only 3 of 12 exceeded 180). Subcutaneous insulin given at 22:00–23:00 has a 6-hour tail and is still active at 03:00–04:00. A repeating cycle is visible: manual insulin 30–60 min after the meal, rescue carbs at 120–180 min, more insulin at 180–210 min. The "🔥 Rage" preset (1.2× insulin, target 80 mg/dL) was used 1–8 times per day, including overnight. (2) **03:00–05:00 is entirely Loop** — 1.32 U of automatic correction at a median BG of 120–126 while asleep. **81% of Loop's overnight automatic insulin is given below 150 mg/dL.** This is the algorithm working as configured: the correction target is a **flat 100 mg/dL for all 24 hours**, so Loop actively corrects from the 120s toward 100 overnight.

5 · CURRENT SETTINGS & QUESTIONS

Basal (38.05 U/day): 00:00 1.45 · 02:00 1.30 · 04:30 1.35 · 09:30 1.80 · 11:00 1.25 · 16:00 1.80 · 17:00 1.70 · 20:00 2.40 · 22:30 2.20

ISF 53 mg/dL/U · **Carb ratio** 6 g/U · **Target** 100 mg/dL · **DIA** 6 h — all flat across the day

Recent totals: bolus 33.9 U/day (down from 43.3 before activity increased) · carbs ~150 g/day

- An overnight correction target above 100?** The target is currently a flat 100 mg/dL for 24 hours, and 81% of Loop's automatic overnight insulin is delivered below 150 — including 1.32 U/night between 03:00 and 05:00 at a median BG of 120–126. Raising the target overnight would reduce the algorithm's own corrections, requires no change in behaviour, and may act more directly on the lows than the basal change. *(Possibly the highest-yield single item here.)*
- The overnight basal rates (22:30 / 00:00 / 02:00)** — is the proposed reduction appropriate, and is the magnitude right?
- Confirm leaving 04:30–09:30 at 1.35.** Loop accepts this block and it produces no lows; we would prefer not to disturb it.
- A rule for late-evening manual corrections** — e.g. no correction below a set threshold after a certain hour, given 60% of manual post-meal doses go in below 150 and 30% below 120.
- Inhaled insulin (Afrezza) for late corrections instead of subcutaneous?** Its ~90-minute action would avoid the overnight tail. The family already uses it; it is not currently logged, so it is not visible in this data.
- Daytime basal.** The 09:30 and 16:00 segments (both 1.80 U/hr) are also well above delivery (Loop suspends 24–62% there). Deliberately excluded here to change one thing at a time — worth revisiting later.
- ISF and carb ratio** are flat all day and unchanged since before the activity increase.

6 · LIMITATIONS

Sample. Marching band began ~Aug 10 and brought both more activity and early wake times. Days with an early start (first meal before 9am, n=7) ran 4.0% below 54 mg/dL versus 0.9% on late-start days (n=52). However those early days were **band camp plus one Saturday event — not the ordinary semester**, where first class will be around 9am. Activity-specific conclusions are therefore not yet reliable and should be re-derived after a few weeks of routine data.

What is not schedule-dependent is the gap between scheduled and delivered basal in \$1 — it holds on early days (20.4 U/day delivered), late days (24.5 U/day), and was present before the move.

Other. Inhaled insulin doses are not logged and are invisible here. Small carbohydrate entries are sometimes used to prompt Loop to correct, so carb totals overstate food intake. G7 sensor-compression artifacts (brief untreated dips that rebound spontaneously) were excluded from glucose statistics; they are frequent overnight and would otherwise inflate the low percentages.